PS#4: Integration by parts - Answer key

1. ()Exercise 5.4.8.15) For each of the following indefinite integrals, determine whether you would use u-substitution, integration by parts, neither, or both to evaluate the integral. In each case, write one sentence to explain your reasoning, and include a statement of any substitutions used. Finally, use your chosen approach to evaluate each integral.

   (a) $\int x^2 \cos(x^3)\, dx = \frac{1}{3}\sin(x^3) + C$

   (b) $\int x^5 \cos(x^3)\, dx = \int x^3 \cdot (x^2 \cos(x^3))\, dx = \frac{1}{3}x^3 \sin(x^3) + \frac{1}{3}\cos(x^3) + C$

   (c) $\int x \ln(x^2)\, dx = \frac{1}{2}\left(x^2 \ln(x^2) - x^2\right) + C$

   (d) $\int \sin(x^4)\, dx$: No elementary antiderivative!

   (e) $\int x^3 \sin(x^4)\, dx = -\frac{1}{4}\cos(x^4) + C$

   (f) $\int x^7 \sin(x^4)\, dx = \int x^4 \cdot (x^3 \sin(x^4))\, dx = -\frac{1}{4}x^4 \cos(x^4) + \frac{1}{4}\sin(x^4) + C$